

On an inflation map for p -valued p -adic Lie groups

Casey Perdue

University of California, San Diego

Introduction

In this poster, we briefly discuss some basic concepts involved in the study of p -adic Lie groups. We generalize a result of Quillen which gives a spectral sequence computing the group cohomology of G/G^p for G a saturated group. We present a morphism of spectral sequences converging to the inflation map from G/G^p to G , then study its consequences for pro- p Iwahori subgroups and unipotent radicals of parabolic subgroups of p -adic reductive groups.

Definitions and generalities

The following definitions involving p -valuations can be found in [6], and the definition of a restricted Lie algebra can be found in [3].

Let G be a pro- p group. A **p -valuation** on G is a function $\omega : G \setminus \{1\} \rightarrow (0, \infty)$ satisfying for any $g, h \in G$:

1. $\omega(g) > \frac{1}{p-1}$,
2. $\omega(g^{-1}h) \geq \min\{\omega(g), \omega(h)\}$,
3. $\omega([g, h]) \geq \omega(g) + \omega(h)$,
4. $\omega(g^p) = \omega(g) + 1$.

We also assume that the subgroups $G_v = \{g \in G \mid \omega(g) \geq v\}$ and $G_{v+} = \{g \in G \mid \omega(g) > v\}$ are open in G , and that G has finite rank.

Given a p -valued pro- p group (G, ω) , one constructs an $\mathbb{F}_p[\pi]$ -Lie algebra

$$\mathrm{gr}G = \bigoplus_{v>0} G_v/G_{v+},$$

where π acts via

$$\pi(gG_{v+}) = g^p G_{(v+1)+} \in G_{v+1}/G_{(v+1)+}.$$

We then get an $\mathbb{F}_p$ -Lie algebra $\mathfrak{g}$ of dimension $\mathrm{rank}(G)$ via

$$\mathfrak{g} = \mathrm{gr}G/\pi(\mathrm{gr}G).$$

We say that (G, ω) is **saturated** if

$$\omega(g) > \frac{p}{p-1} \implies g = h^p \text{ for some } h \in G.$$

If (G, ω) is saturated, then

$$G^p = \{g^p \mid g \in G\}$$

is an open normal subgroup of index $p^{\mathrm{rank}(G)}$, and G/G^p is naturally endowed with a quotient filtration via ω .

An $\mathbb{F}_p$ -Lie algebra is **restricted** if it is endowed with a p -mapping $\mathfrak{g} \rightarrow \mathfrak{g}$, $X \mapsto X^{[p]}$, satisfying for any $X, Y \in \mathfrak{g}$ and $t \in \mathbb{F}_p$:

1. $\mathrm{ad}(X^{[p]}) = (\mathrm{ad}X)^p$,
2. $(tX)^{[p]} = tX^{[p]}$,
3. $(X+Y)^{[p]} = X^{[p]} + Y^{[p]} + \sum_{i=1}^{p-1} s_i(X, Y)$, where $i \cdot s_i(a, b)$ is the coefficient of λ^{i-1} in $(\mathrm{ad}(\lambda X + Y))^{p-1}(X)$.

Given a restricted $\mathbb{F}_p$ -Lie algebra $\mathfrak{g}$, its **restricted universal enveloping algebra** is

$$U_{\mathrm{res}}(\mathfrak{g}) = U(\mathfrak{g})/\langle X^p - X^{[p]} \mid X \in \mathfrak{g} \rangle,$$

and its **restricted Lie algebra cohomology** is $H_{\mathrm{res}}^i(\mathfrak{g}, -) = H^i(U_{\mathrm{res}}(\mathfrak{g}), -)$.

The spectral sequences

Let (G, ω) be a saturated group. Using a slight generalization of a result proved by Quillen in [5], one can show that the Lazard Lie algebra $\mathfrak{g} = \mathrm{gr}G/\pi(\mathrm{gr}G)$ is a restricted Lie algebra with zero p -mapping $X^{[p]} = 0$. One can then produce using a similar argument to that in [1] an E_1 -spectral sequence which Day calls the May spectral sequence:

$$H_{\mathrm{res}}^{s,t}(\mathfrak{g}, \mathbb{F}_p) \implies H^{s+t}(G/G^p, \mathbb{F}_p).$$

By [7], there is also an E_1 -spectral sequence

$$H_{\mathrm{cl}}^{s,t}(\mathfrak{g}, \mathbb{F}_p) \implies H^{s+t}(G, \mathbb{F}_p),$$

where H_{cl} is the classical Lie algebra cohomology. The inflation map induces a morphism of spectral sequences

$$\begin{array}{ccc} H_{\mathrm{res}}^{s,t}(\mathfrak{g}, \mathbb{F}_p) & \implies & H^{s+t}(G/G^p, \mathbb{F}_p) \\ \downarrow & & \downarrow \mathrm{inf} \\ H_{\mathrm{cl}}^{s,t}(\mathfrak{g}, \mathbb{F}_p) & \implies & H^{s+t}(G, \mathbb{F}_p) \end{array}$$

where the maps on the left come from the canonical map

$$H_{\mathrm{res}}^i(\mathfrak{g}, -) \rightarrow H_{\mathrm{cl}}^i(\mathfrak{g}, -).$$

Key examples: pro- p Iwahori and unipotent radical of parabolic

Let $\mathbf{G}$ be a connected reductive group defined over a p -adic field K which splits over an extension L/K . Then we have two main pro- p groups of interest:

- Let $\mathbf{P} \leq \mathbf{G}$ be a parabolic K -subgroup corresponding to a set $J \subseteq \Phi^+$ of simple roots, $\mathbf{U} = \prod_{\alpha \in \Phi^+, \mathrm{nd} \setminus \Phi_J^+, \mathrm{nd}} U_\alpha \leq \mathbf{P}$ its unipotent radical. Let x_0 denote an absolutely special point in the Bruhat-Tits building for G . We then seek to consider

$$U = \prod_{\alpha \in \Phi^+, \mathrm{nd} \setminus \Phi_J^+, \mathrm{nd}} U_{\alpha, x_0, 0} \leq \mathbf{U}(K).$$

- There is a pro- p Sylow subgroup $I \leq \mathbf{G}(K)$ called the **pro- p Iwahori** subgroup which contains U .

Let $e = e(L/\mathbb{Q}_p)$ be the absolute ramification index, and let h be the Coxeter number of G (the height of the highest root plus one).

If $p-1 > he$, then both I and U are saturated with respect to a natural p -valuation ω . Thus, $U^p \leq U$ and $I^p \leq I$ are normal subgroups.

For simplicity, assume $\mathbf{G} = \mathrm{SL}_n$ is defined over $\mathbb{Q}_p$ and $\mathbf{P}$ is the standard Borel, in which case we are assuming that $p-1 > n$, and we can write

$$U = \left\{ \begin{pmatrix} 1 & * & \\ & \ddots & \\ 0 & & 1 \end{pmatrix} \in \mathrm{SL}_n(\mathbb{Z}_p) \right\}, \quad I = \left\{ g \in \mathrm{SL}_n(\mathbb{Z}_p) \mid g \equiv \begin{pmatrix} 1 & * & \\ & \ddots & \\ 0 & & 1 \end{pmatrix} \pmod{p} \right\}.$$

We first note that

$$U/U^p \cong U_n(\mathbb{F}_p) = \left\{ \begin{pmatrix} 1 & * & \\ & \ddots & \\ 0 & & 1 \end{pmatrix} \in \mathrm{SL}_n(\mathbb{F}_p) \right\}.$$

As for I/I^p , there is a normal elementary abelian p -subgroup $K \leq I/I^p$ such that

$$I/I^p \cong K \rtimes U_n(\mathbb{F}_p),$$

from which we can produce a natural faithful $\frac{1}{2}(n^2 + n)$ -dimensional representation over $\mathbb{F}_p$. For example, if $n = 2$, this representation corresponds to an isomorphism $I/I^p \cong U_3(\mathbb{F}_p)$.

Input from Lie algebra cohomology

One can show that the p -valuation ω creates Lazard Lie algebras $\mathfrak{u}$ and $\mathfrak{i}$ of U and I , respectively, which are isomorphic to the Lie algebras coming from the filtrations on U/U^p and I/I^p coming from the filtration induced by the lower central series on these finite groups. Moreover, tensoring these up to $\overline{\mathbb{F}_p}$ gives the Lie algebras which come from the algebraic groups U_n for U/U^p , and U_3 for I/I^p when $n = 2$. Using [2, Proposition 3.5], one can then deduce the following:

The canonical maps

$$H_{\mathrm{res}}^*(\mathfrak{u}, \mathbb{F}_p) \rightarrow H_{\mathrm{cl}}^*(\mathfrak{u}, \mathbb{F}_p)$$

are always surjective, and if $n = 2$, the following canonical map is also surjective:

$$H_{\mathrm{res}}^*(\mathfrak{i}, \mathbb{F}_p) \rightarrow H_{\mathrm{cl}}^*(\mathfrak{i}, \mathbb{F}_p).$$

Example application

Consider either the case of $U = U_n(\mathbb{Z}_p)$ for $\frac{1}{6}(n^3 - n) \leq p$ or the pro- p Iwahori $I \leq \mathrm{SL}_2(\mathbb{Z}_p)$.

As an example application of these ideas, one can deduce from certain computations performed in [4] and [1] that for each nonzero element on the E_1 -pages of the spectral sequences

$$H_{\mathrm{cl}}^{s,t}(\mathfrak{u}, \mathbb{F}_p) \implies H^{s+t}(U, \mathbb{F}_p) \quad \text{and} \quad H_{\mathrm{cl}}^{s,t}(\mathfrak{i}, \mathbb{F}_p) \implies H^{s+t}(I, \mathbb{F}_p),$$

their lifts under the surjective graded canonical maps from restricted Lie algebra cohomology are killed under every differential in the top spectral sequence. Thus, an easy inductive argument yields that the maps in our morphism of spectral sequences are still surjective on the E_∞ -pages, so the graded inflation maps

$$\mathrm{gr}H^*(U/U^p, \mathbb{F}_p) \rightarrow \mathrm{gr}H^*(U, \mathbb{F}_p), \quad \mathrm{gr}H^*(I/I^p, \mathbb{F}_p) \rightarrow \mathrm{gr}H^*(I, \mathbb{F}_p)$$

are surjective. In particular, this shows that the (ungraded) inflation maps

$$H^*(U/U^p, \mathbb{F}_p) \rightarrow H^*(U, \mathbb{F}_p), \quad H^*(I/I^p, \mathbb{F}_p) \rightarrow H^*(I, \mathbb{F}_p)$$

are also surjective in our two key cases $U = U_n(\mathbb{Z}_p)$ for $\frac{1}{6}(n^3 - n) \leq p$, and $I \leq \mathrm{SL}_2(\mathbb{Z}_p)$ pro- p Iwahori.

References

- [1] Paul Julian Day. The may spectral sequence, 1994.
- [2] Eric M Friedlander and Brian J Parshall. Geometry of p -unipotent lie algebras. *Journal of Algebra*, 109(1):25–45, 1987.
- [3] Nathan Jacobson. *Lie Algebras*. Dover Books on Mathematics. Dover Publications, 1st rev. edition, 1979.
- [4] Daniel Kongsgaard. On the mod p cohomology of pro- p iwahori subgroups of $\mathrm{SL}_n(\mathfrak{o}_p)$, 2022.
- [5] Daniel G. Quillen. On the associated graded ring of a group ring. *Journal of Algebra*, 10(4):411–418, 1968.
- [6] Peter Schneider. *p-Adic Lie Groups*, volume 344 of *Grundlehren der mathematischen Wissenschaften*. Springer, 1st edition, 2011.
- [7] Claus Sorensen. Hochschild cohomology and p -adic lie groups. 2019.